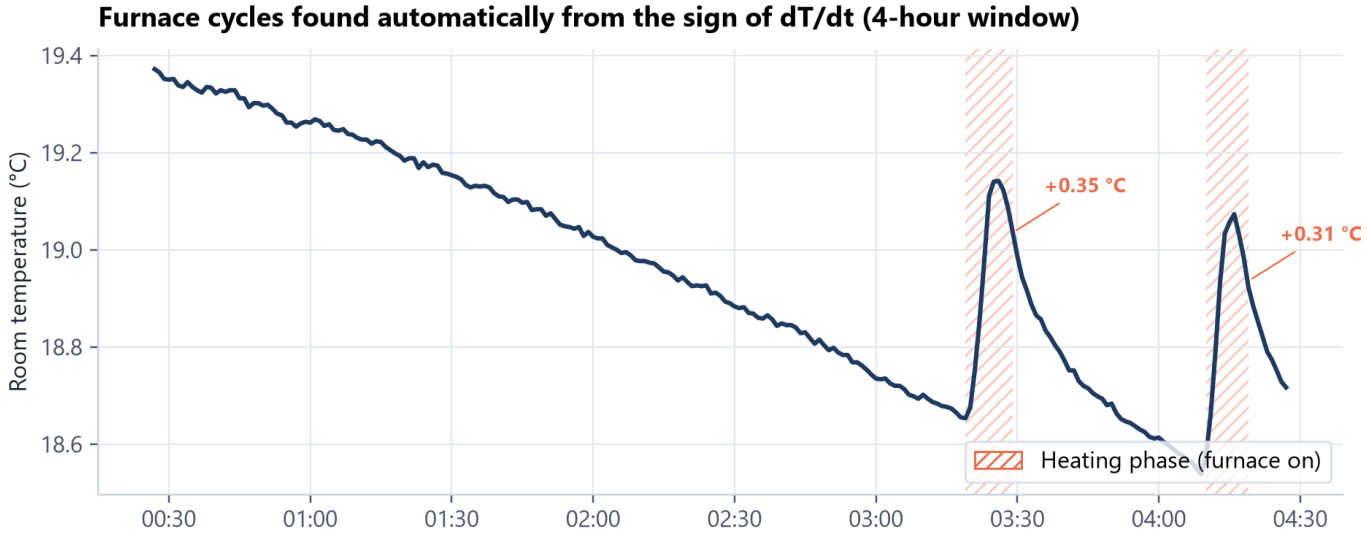
PROJECT PORTFOLIO · 2026

Why is one room colder?

Measuring heat loss with ESP32 sensor nodes, weather data and an RC circuit model. Plus: a closed-loop LED lamp.

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Real data: Room A, Oct 3 to 4, 2026. 5,747 sensor readings; the furnace's heating sawtooth, detected automatically.

THE QUESTION

Three explanations, three different fixes

One bedroom is always colder. Before spending money on a fix, find out **which** of these is actually true.

H1

Less sun

Window faces away from the sun

DATA WOULD SHOW

Gap in daytime only; disappears at night

FIX

Curtains or accept

H2

Leaky envelope

Poor insulation, air leaks (low R)

DATA WOULD SHOW

Room cools faster after every furnace cycle

FIX

Weatherstrip, caulk, insulate

H3

Weak heating

Damper partly closed, long duct

DATA WOULD SHOW

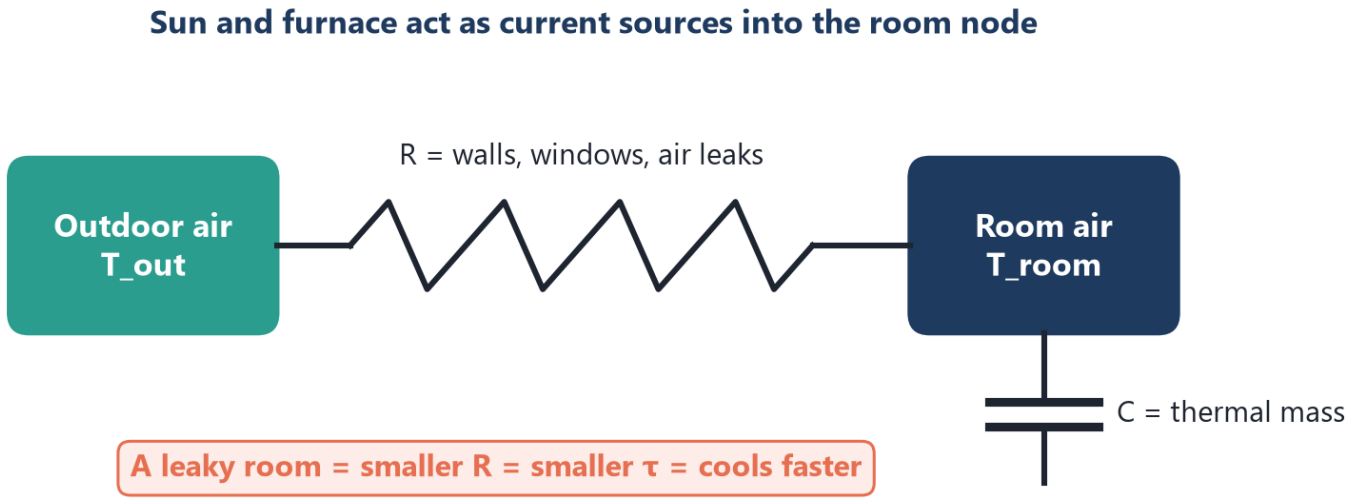
Room warms less per furnace cycle

FIX

Open or balance dampers

THE IDEA

A room behaves like an RC circuit



$$C \cdot dT/dt = (Tout - T) / R + a \cdot solar + b \cdot furnace$$

- Walls and air leaks = **resistor R**. Heat flow ∝ temperature difference, like Ohm's law.
- Air and furniture = **capacitor C**: they store heat.
- **τ = RC** is one number per room. A leaky room has smaller R, smaller τ, and cools faster.
- Comparing τ between rooms separates H2 from H1 and H3.

APPARATUS

How it's measured

Environmental monitor: sensing to analysis



10 s

sampling interval, temperature · RH · pressure · light

±0.2 °C

target agreement between nodes after calibration

1 min

resampled grid, joined to hourly Edmonton weather

100 %

of derived values reproducible as Excel formulas

ESP32 · Bosch BME280 · ROHM BH1750 · Arduino C++ · Python (pandas, scikit-learn) · Open-Meteo API · Microsoft Excel

METHOD

The experiment, phase by phase

0

Calibrate

Both nodes side by side for 24 h. Measure the offset, subtract it everywhere.

Pending

1

Baseline

One node in Room A. Build and test the whole pipeline on real data.

Done

2

Two rooms

One node per room, 1.2 m high, away from vents. Log events.

2 to 3 weeks

3

Analyse

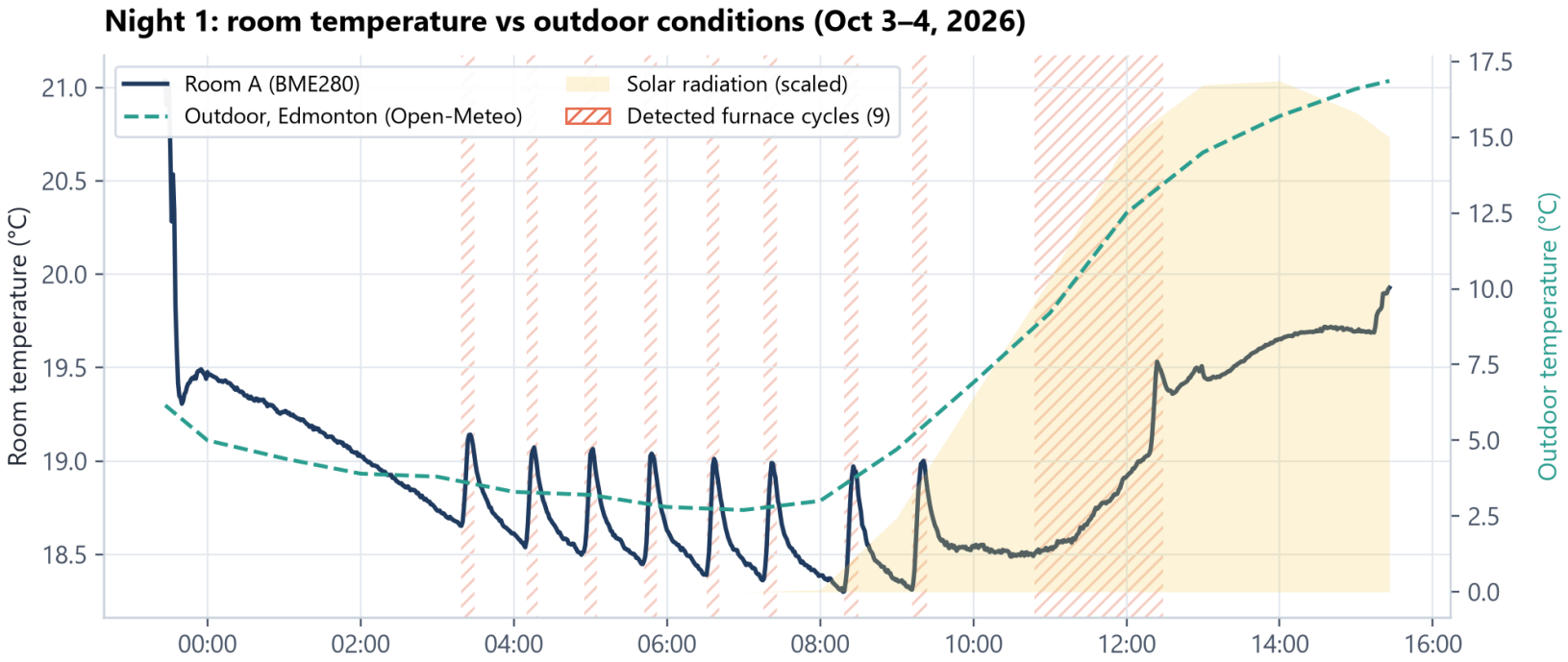
τ per room, rise per furnace cycle, day vs night, wind.

Pipeline ready

Why calibrate first: each sensor is only ±1 °C accurate out of the box, which is bigger than the effect being measured.

RESULTS · MEASURED

Night 1: the room, the weather and the furnace



The thermostat holds about 19 °C while it drops to 2.7 °C outside.
τ ≈ 24 h: with the furnace off, Room A would lose about 0.8 °C in the first hour.

18.9 °C

average room temperature

2.7 → 16.9 °C

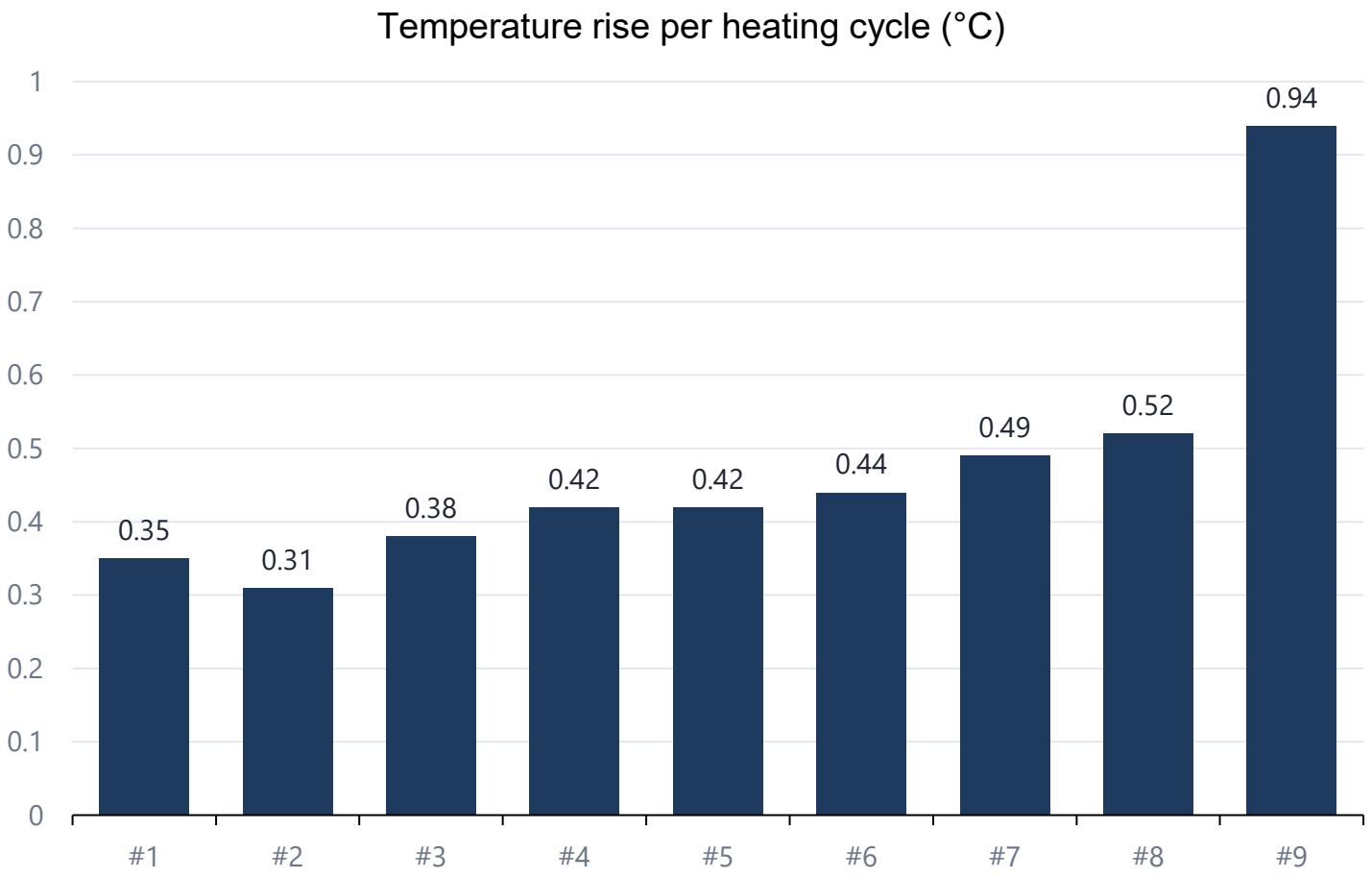
outdoor range

23.9 h

Room A time constant τ (RC fit)

RESULTS · MEASURED

9 furnace cycles, found automatically



0.48 °C

average rise per cycle

56 min

average time between cycles

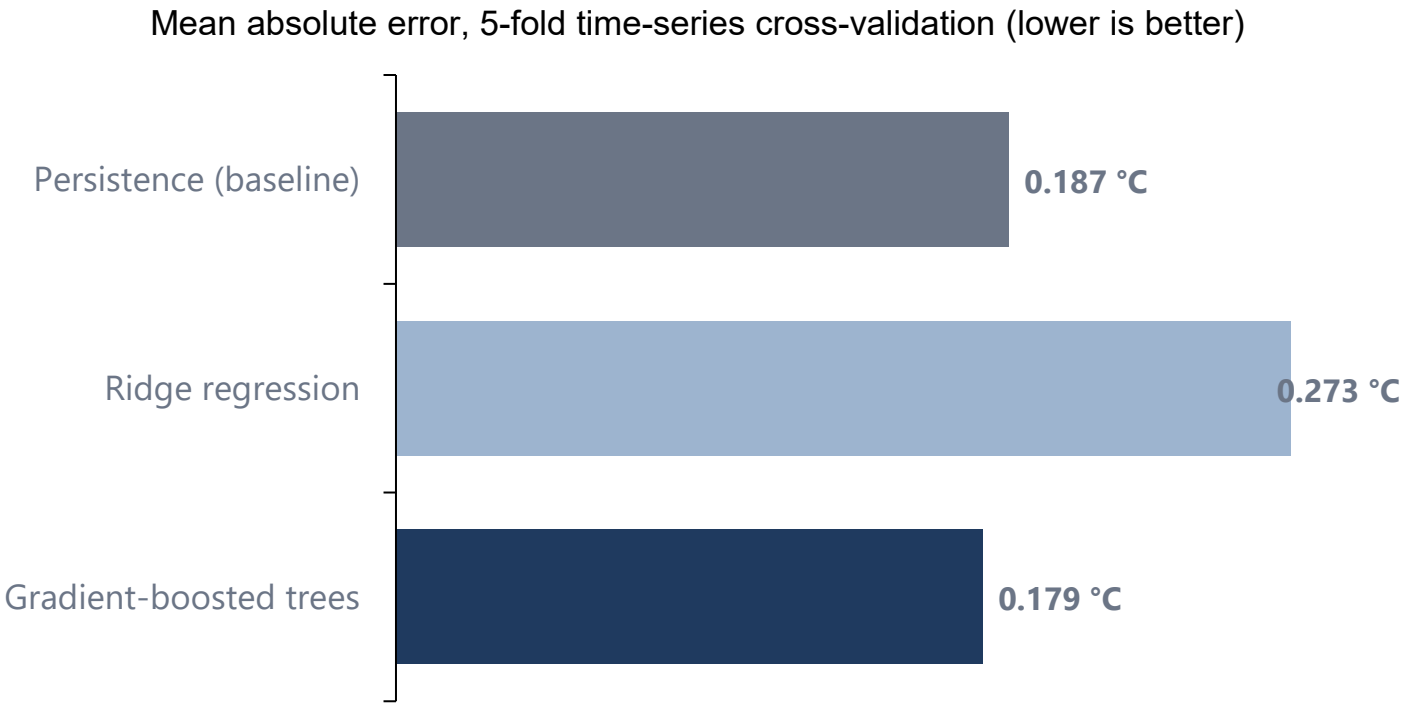
≈ 20 min

furnace on-time per cycle

Method: smooth over 5 min, find where dT/dt changes sign, keep rises ≥ 0.15 °C (15× the sensor resolution).

EXTENSION · MEASURED

Can ML forecast the room 30 minutes ahead?



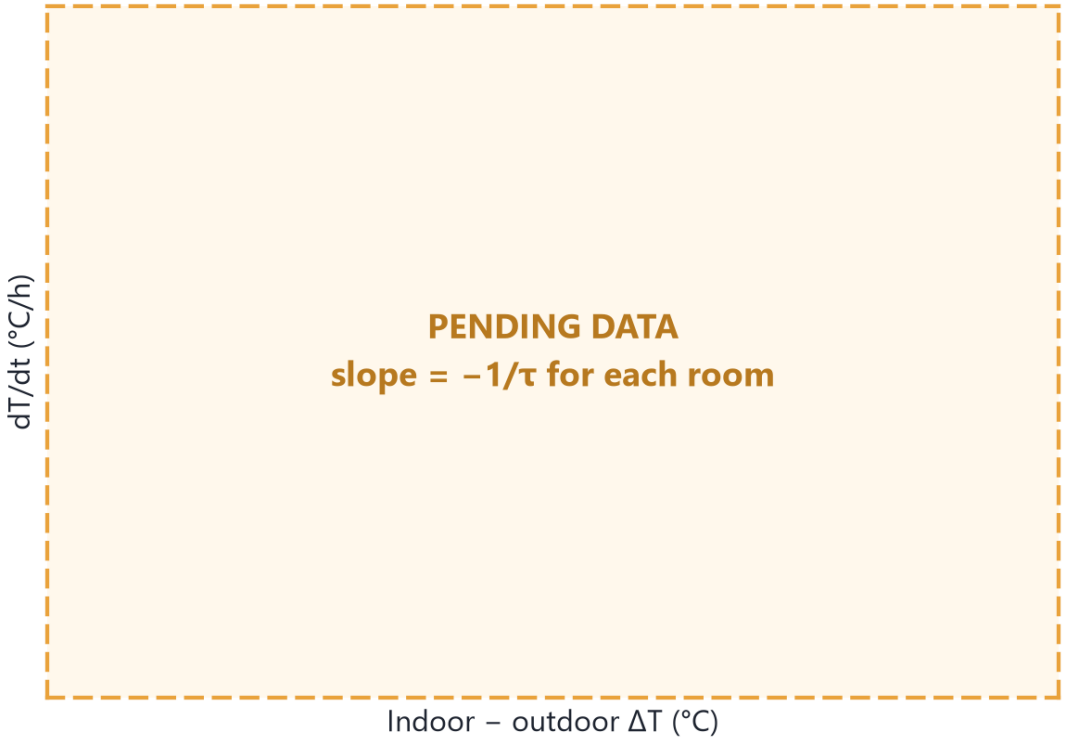
Features: recent temperature changes, indoor–outdoor gap, solar radiation. Python + scikit-learn.

HONEST RESULT

- Best model beats "no change" by only **4.7 %**, and loses on a single hold-out split.
- One night holds just 9 furnace cycles. The model needs weeks of data.
- Tested so it never trains on the future (time-series folds).

Room A vs Room B: the actual experiment

Heat-loss signature: cooling rate vs ΔT, furnace off



PENDING DATA

What gets computed per room

Metric	Room A	Room B
Time constant τ (h)	23.9	?
Rise per furnace cycle (°C)	0.48	?
Cooling rate per °C ΔT (1/h)	0.063	?
Day − night gap (°C)	?	?

Every chart, table and formula is already built. Room B's logs get pasted into the workbook and the results fill in.

INTERPRETATION

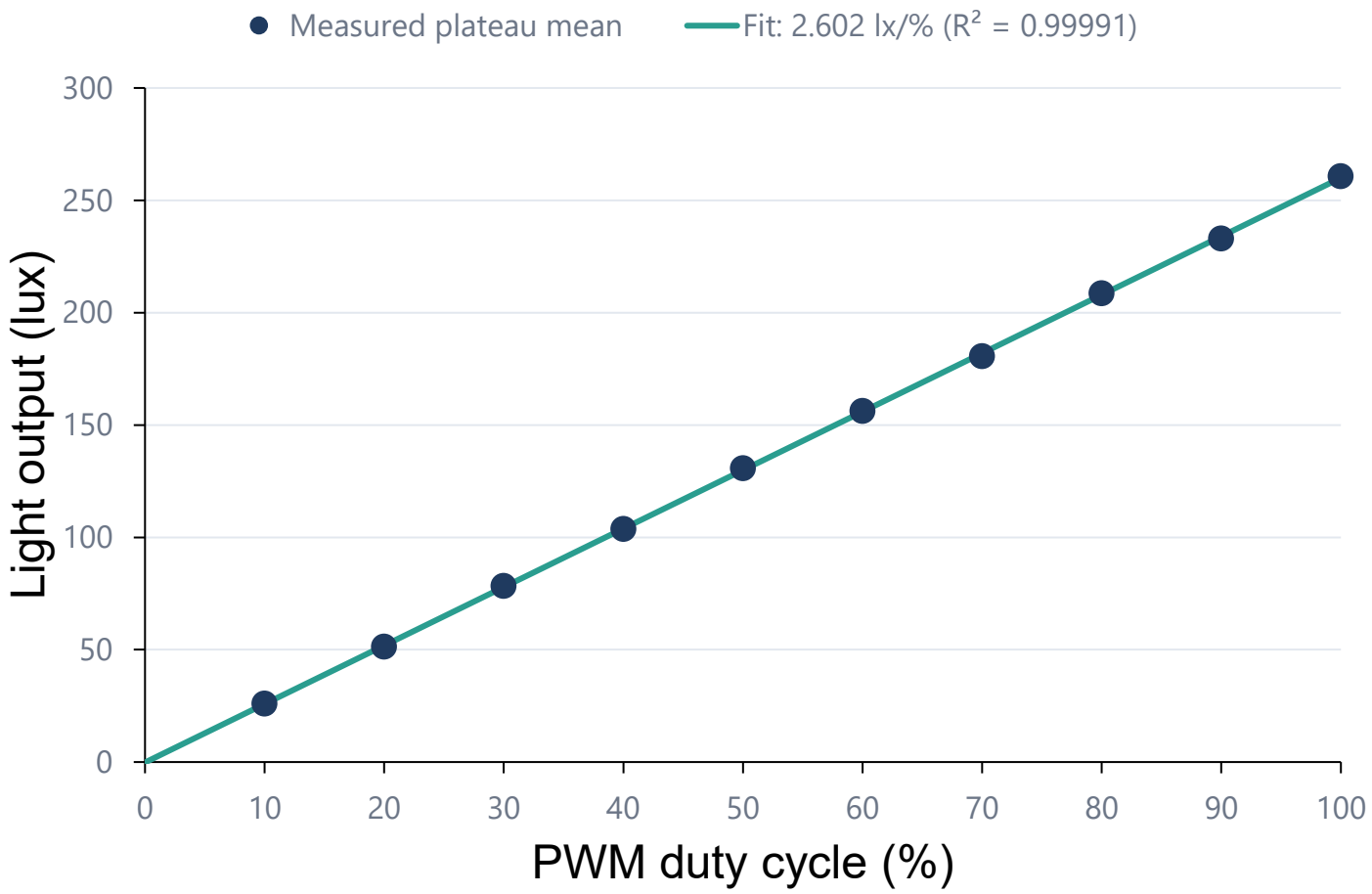
Decision rules, written before the data arrives

τ(B) at least 25 % smaller than τ(A), day and night	→	**H2 · leaky envelope** Weatherstrip, seal outlets on exterior walls
Similar τ, but B warms clearly less per cycle	→	**H3 · weak heating** Open or balance dampers, check the duct
Gap mostly in daytime; B's light curve peaks late or low	→	**H1 · less sun** Accept, or thermal curtains at night
Gap grows with wind speed	→	**Air leaks (part of H2)** Air-seal first: the cheapest fix

Fixing the rules first stops me from bending the result toward the answer I expected.

ADAPTIVE LED LAMP · MEASURED

Project 2: is the LED lamp's dimming linear?



0.99991

R^2, light vs duty

0.51 %

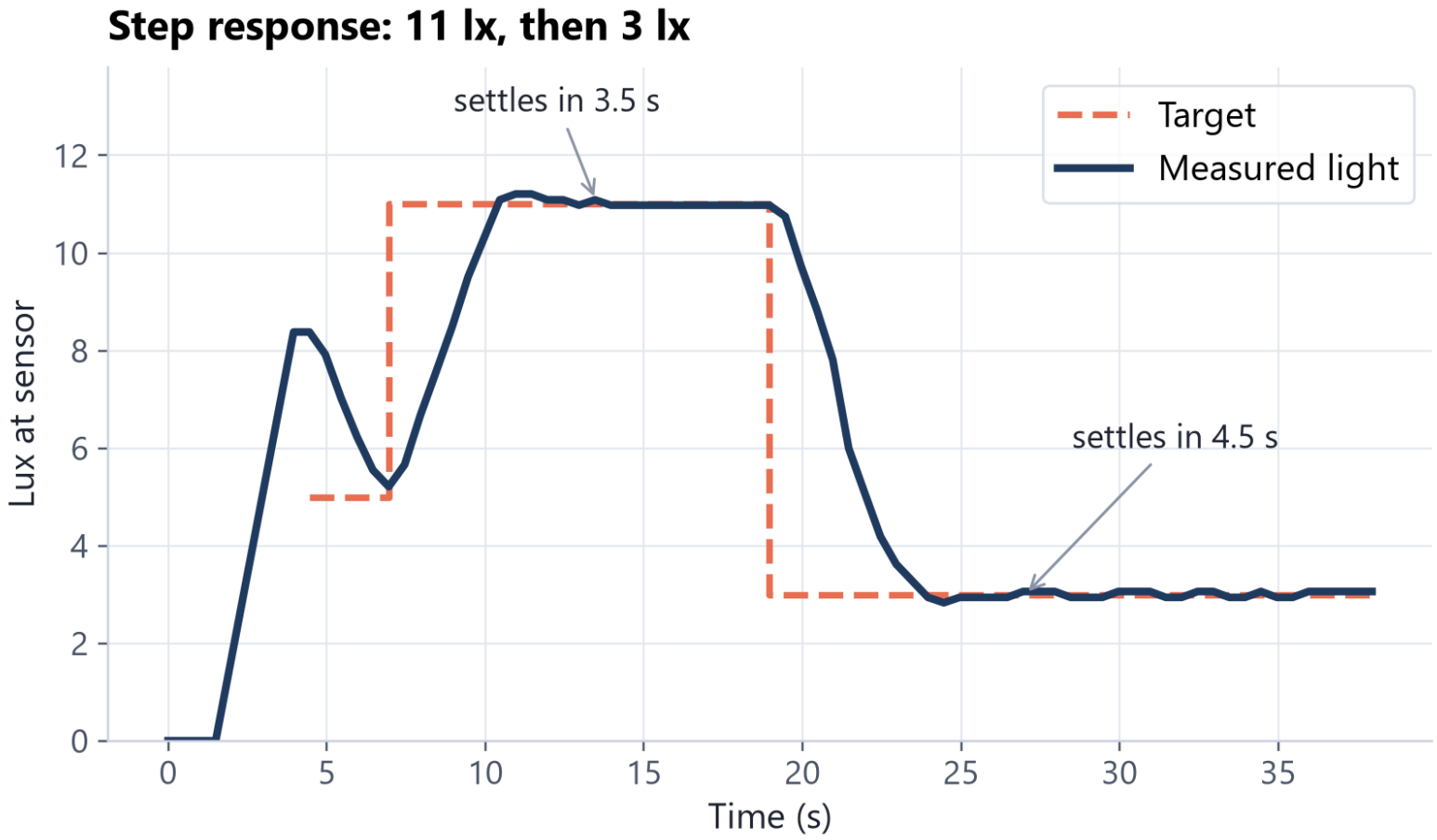
worst deviation from the line (of full scale)

7×

finer sensor resolution after fixing the first failed test

ESP32 · IRLZ44N MOSFET · 12 V strip · BH1750

Holding a set brightness with feedback

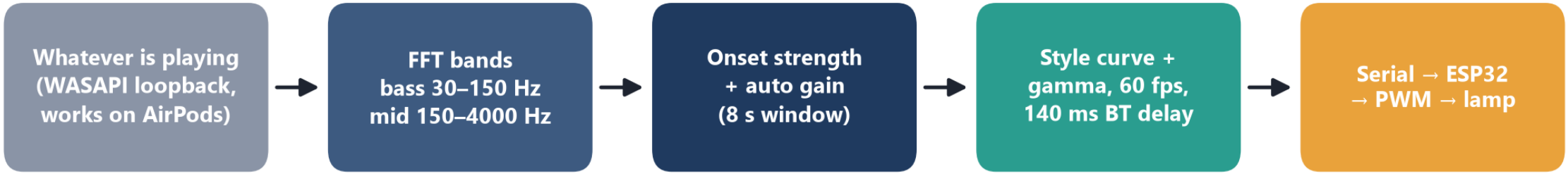


$$u \leftarrow u + (G/K)\cdot error$$

- **Integral controller**, gain G = 0.35
- Plant gain K **measured by the lamp itself**, since it changes with sensor position
- Simulated first: stable even with K wrong by 2×
- **3.5 s** to settle at 11 lx, **0.21 lx** overshoot, ±0.1 lx ripple

Then I made it react to music

Music-reactive mode: from the audio stream to the lamp



0 mics

it listens to the audio stream itself, so headphones work and room noise is ignored

60 fps

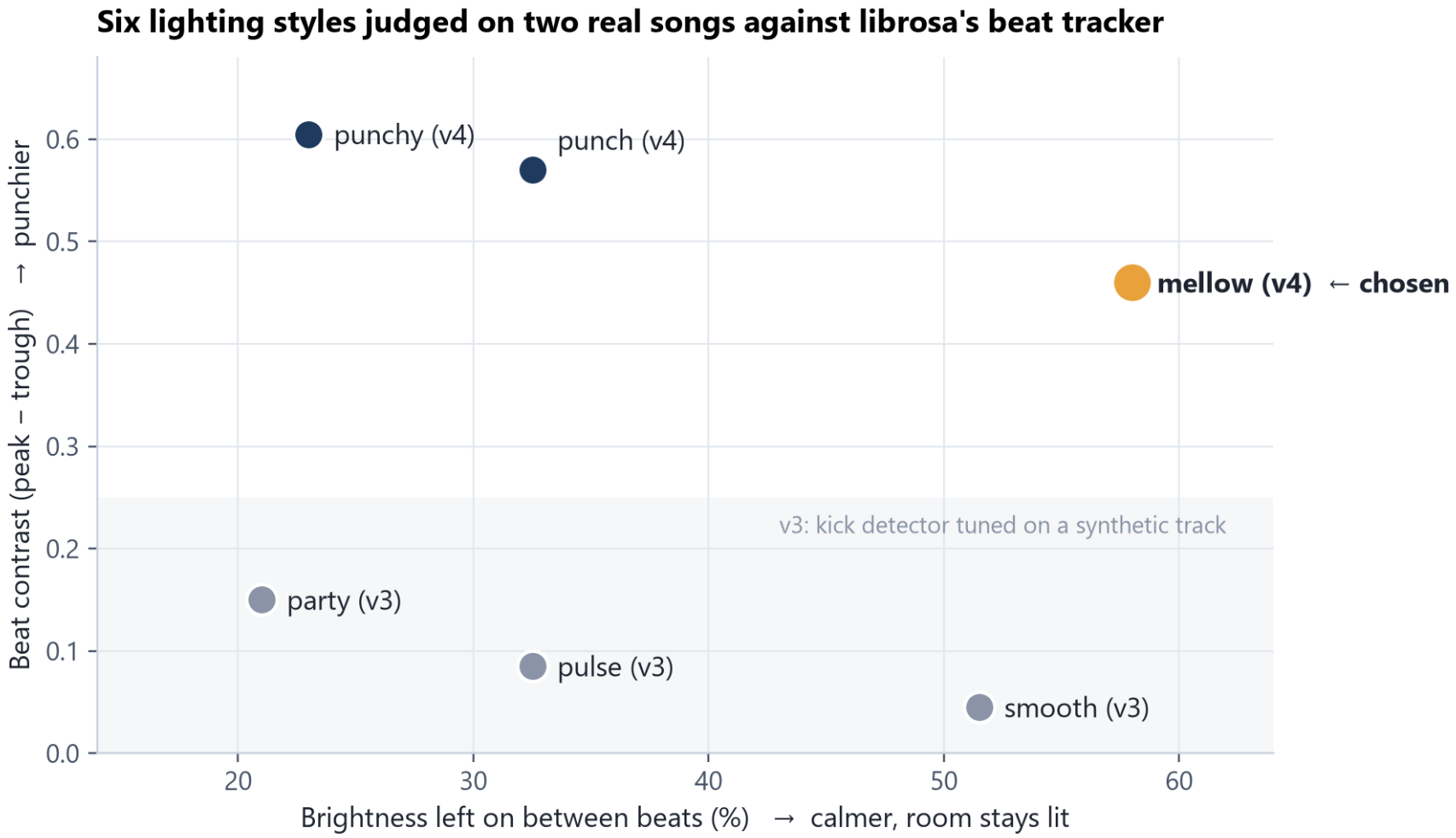
brightness updates, sent over serial to the ESP32's PWM

140 ms

Bluetooth delay measured and compensated, so flashes land on the beat

Same lamp hardware as before. The music mode is a Python engine on the PC: numpy FFT, auto gain control, onset detection.

Four versions, judged against an independent beat tracker



v1 Mapped raw volume. Swung only 50–70 % and followed the volume knob, not the music.

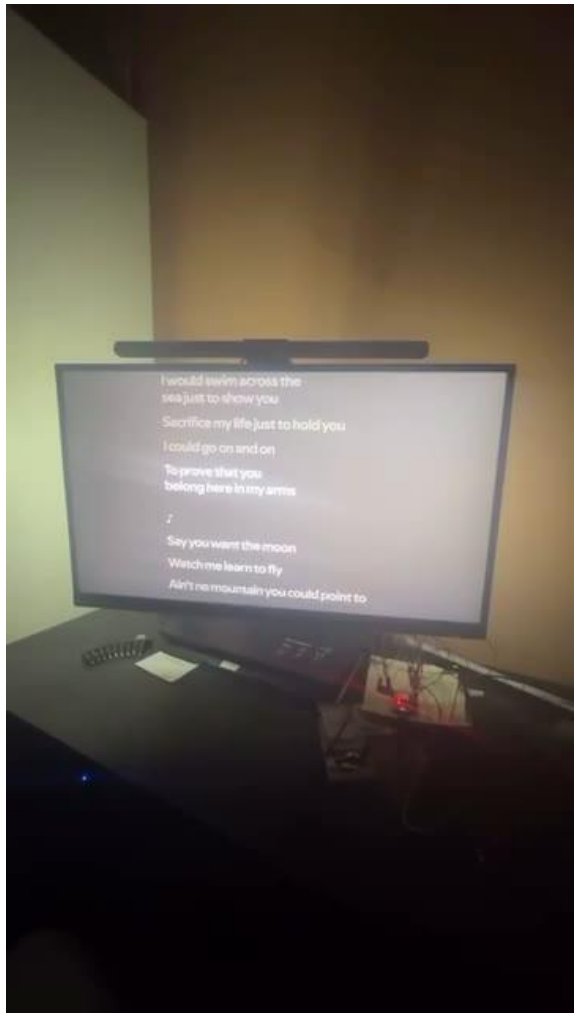
v2 Auto gain over an 8 s window, so it works at any volume.

v3 Kick detector tuned on a synthetic track. On real songs it caught only 0.12–0.33 kicks per beat.

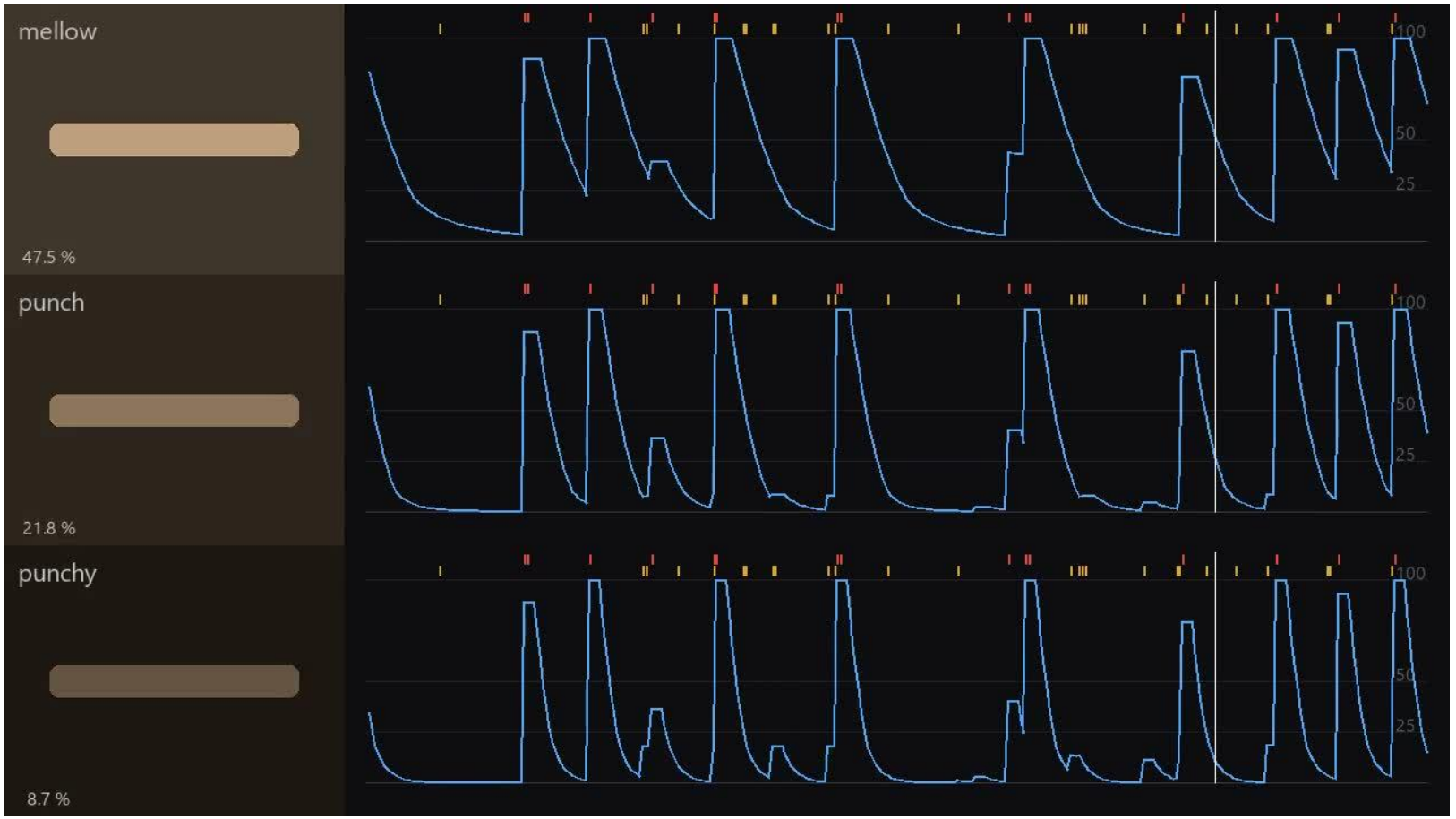
v4 Continuous bass/mid onset strength: 5.8× the beat contrast on the same song.

Chosen: mellow. Picked by watching all three v4 styles on the same song. It keeps the room lit (58 %) and still lands the beats.

Watch it



Real lamp, behind the monitor (click to play)



Rendered comparison: the three v4 styles run on the same 30 s of music. Left: simulated diffuser at the commanded brightness. Right: brightness trace with detected hits.

CONCLUSIONS

What I built, and what's next

Done

- Sensor → ESP32 → Wi-Fi → Python → Excel chain, validated on 16 h of real data
- Room A: 9 furnace cycles, ≈ 0.5 °C each, $\tau \approx 24$ h
- LED lamp: linear to R^2 0.9999, closed-loop control within 1 sensor count
- Music mode: 4 versions, judged against librosa; v4 has 5.8× the beat contrast

Next

- Build node 2 and run the 24 h side-by-side calibration
- Log both rooms for 2 to 3 weeks
- Fill in the comparison and apply the decision rules

The same approach transfers to any building or equipment question: measure, model, then decide.

Thank you

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Technical report (Word)

Analysis workbook (Excel)

Firmware + Python code

available on request